

RAFFLES INSTITUTION

2018 Year 6 Preliminary Examination
Higher 1

BIOLOGY

8876/01

Paper 1 Multiple Choice

25th September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

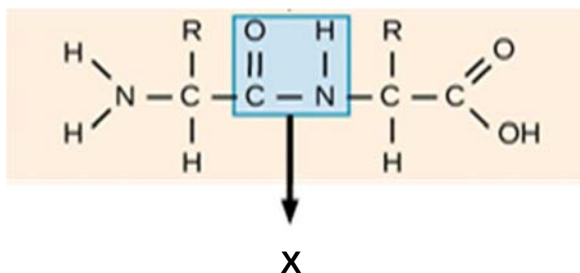
(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **21** printed pages.



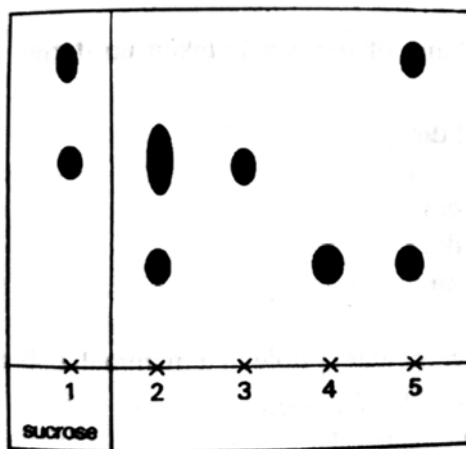
Raffles Institution
Internal Examination

1. The diagram shows a molecular structure.



What is enclosed by the box X?

- A Phosphodiester bond
 B Glycosidic bond
 C Ester bond
 D Peptide bond
2. Five disaccharides were each hydrolysed and the products were separated by chromatography. The chromatogram produced is shown in the diagram.

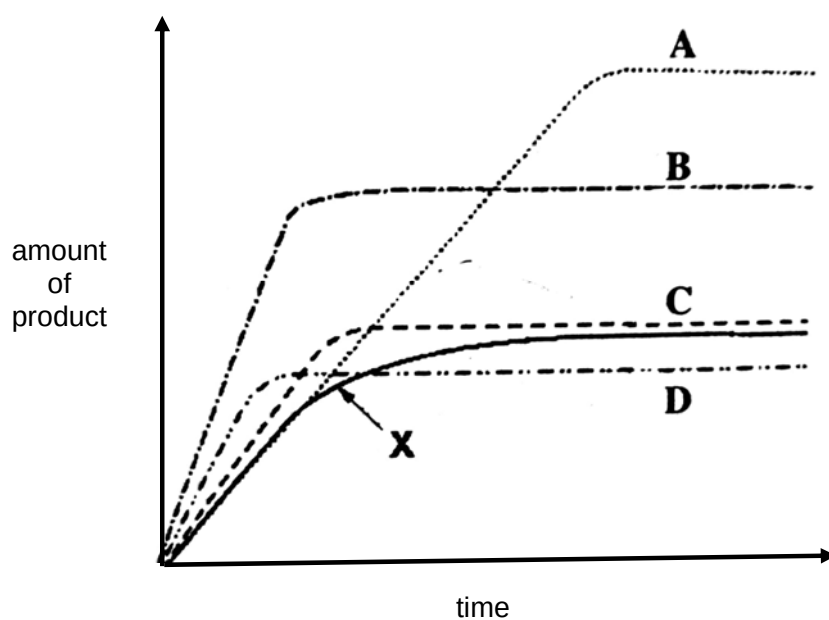


If lane 1 represents the products of the hydrolysis of sucrose, which results were obtained from the hydrolysis of lactose and maltose?

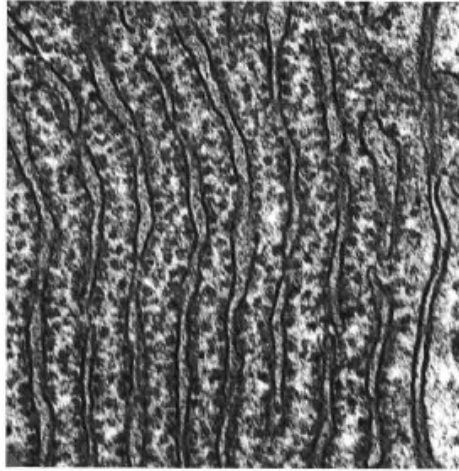
	lactose	maltose
A	2	3
B	2	4
C	5	2
D	5	3

3. An essential feature of a competitive inhibitor is its ability to _____.
- A activate an operator gene
 - B modify a substrate
 - C occupy an active site
 - D same shape as active site
4. Line X shows the activity of an enzyme at 20°C. Lines A to D show the effect of different conditions on the activity of the enzyme.

Which line shows the effect of increasing the temperature by 10°C and adding extra substrate?

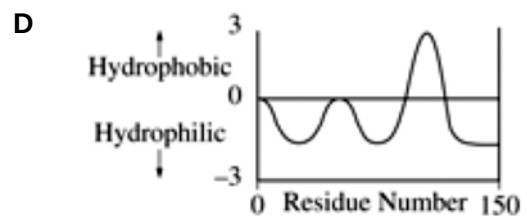
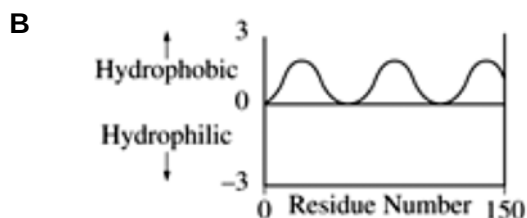
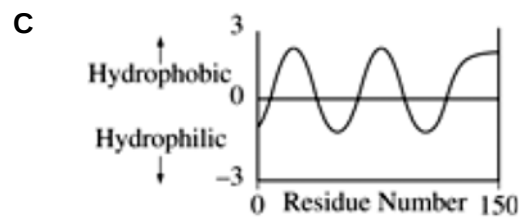
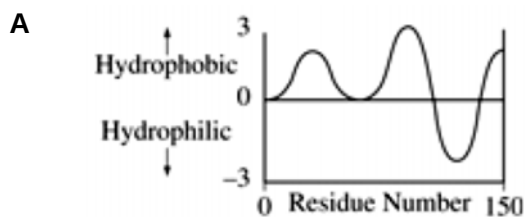


5. The electron micrograph shows part of an organelle in a cell.



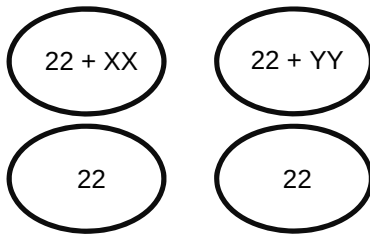
What describes a function of the cisternae in the organelle shown?

- A Moving protein to places where they are covered by phospholipid membranes for secretion outside the cell
 - B Producing proteins and covering them with phospholipid membranes for secretion outside the cell
 - C Producing proteins, covering them with phospholipid membranes and moving them for use inside the cell
 - D Producing ribosomes and proteins and storing them in phospholipid membranes for use inside the cell
6. Glycophorin, an integral membrane protein, has a single transmembrane α helix. Which of the following plots most likely represents glycophorin?

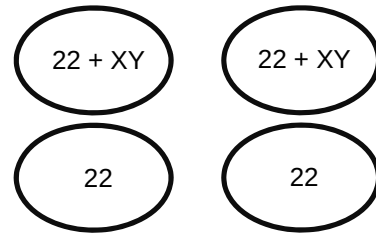


7. A human cell with 44 autosomes, and sex chromosomes X and Y, suffers a non-disjunction at the first meiotic division. Which of the following set of gametes could result?

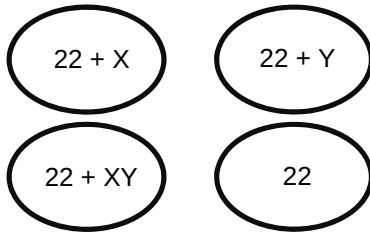
A



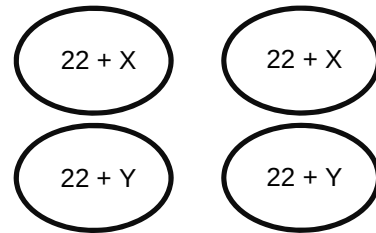
C



B



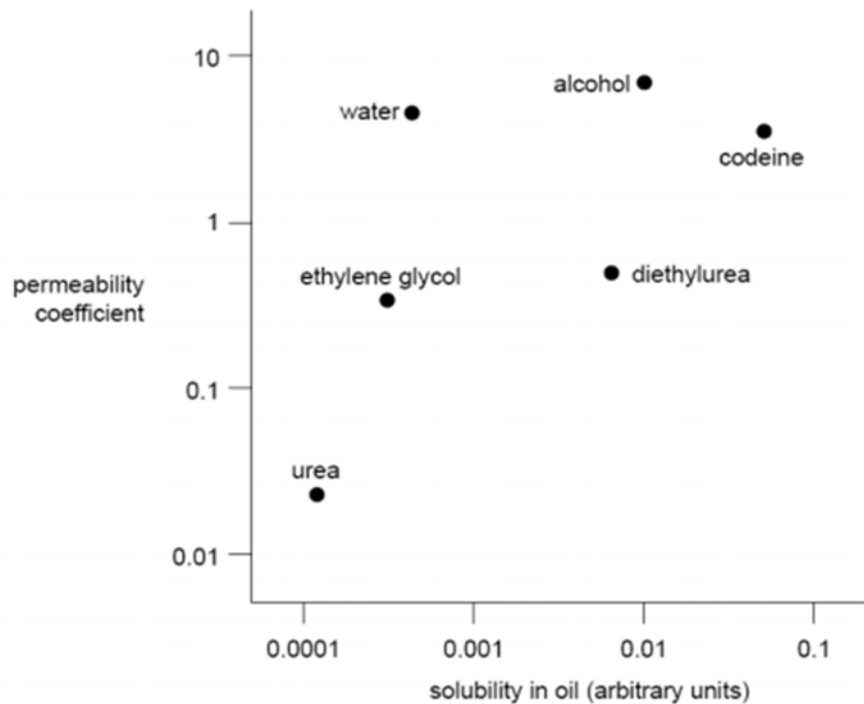
D



8. Which of the following is **not** a function of mitosis?

- A Growth of multicellular organisms.
- B Repair of multicellular organisms.
- C Reproduction of single-celled organisms.
- D Increase in size of single-celled organisms.

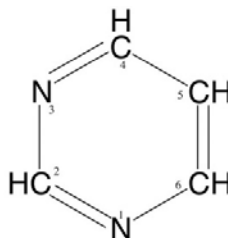
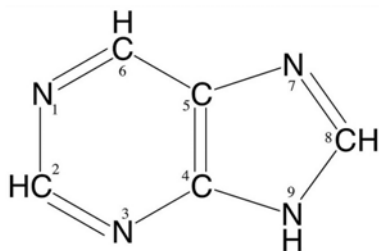
9. Materials are constantly exchanged between a cell and its surroundings. The ease with which a molecule passes through a plasma membrane can be represented by a permeability coefficient. The graph below shows the permeability coefficients for a range of molecules plotted against their solubility in lipid.



From the graph you can conclude that

- A water is more lipid-soluble than alcohol.
 - B a cell gets rid of urea more slowly than excess water.
 - C drugs like alcohol and codeine enter cells more slowly than ethylene glycol.
 - D ethylene glycol passes through the plasma membrane more easily than diethylurea.
10. Which of the following statements is **not** true of mutations in a diploid organism?
- A Chances of encountering a recessive lethal allele in a population of diploid organism is higher compared to that of haploid organisms.
 - B All somatic mutations alter the organism's phenotypes.
 - C Some mutations causes the loss in regulation of the cell cycle.
 - D Some loci are more prone to mutation than others.

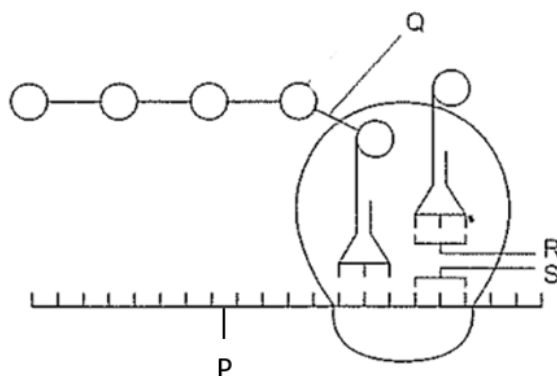
11. The structures of purine and pyrimidine are shown below.



Which of the following correctly shows the numbers of carbon atoms in the corresponding nucleic acid?

	Molecule	Number of carbons in the molecule
A	DNA strand with the sequence ATCGAAA	33
B	mRNA molecule with the sequence AUCGAAA on 1 strand	30
C	DNA molecule with the sequence ATCGAAA on 1 strand	33
D	DNA strand with the sequence ATCGAAA	68

12. The diagram below shows the process of translation in a prokaryotic cell.



Which of the following correctly identifies the bonds?

	P	Q	between R and S
A	Peptide	Phosphodiester	Hydrogen
B	Hydrogen	Disulfide	Phosphodiester
C	Phosphodiester	Peptide	Hydrogen
D	Hydrogen	Hydrogen	Peptide

13. The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenised cell contents through a fractionating column. The column has short lengths of uracil nucleotides attached to a solid supporting material. Molecules of mRNA that can pass through the column are quickly broken up into small pieces and cannot be translated.

The active mRNAs that attach to the column can be collected subsequently by an appropriate treatment.

Which statements correctly describe active mRNA?

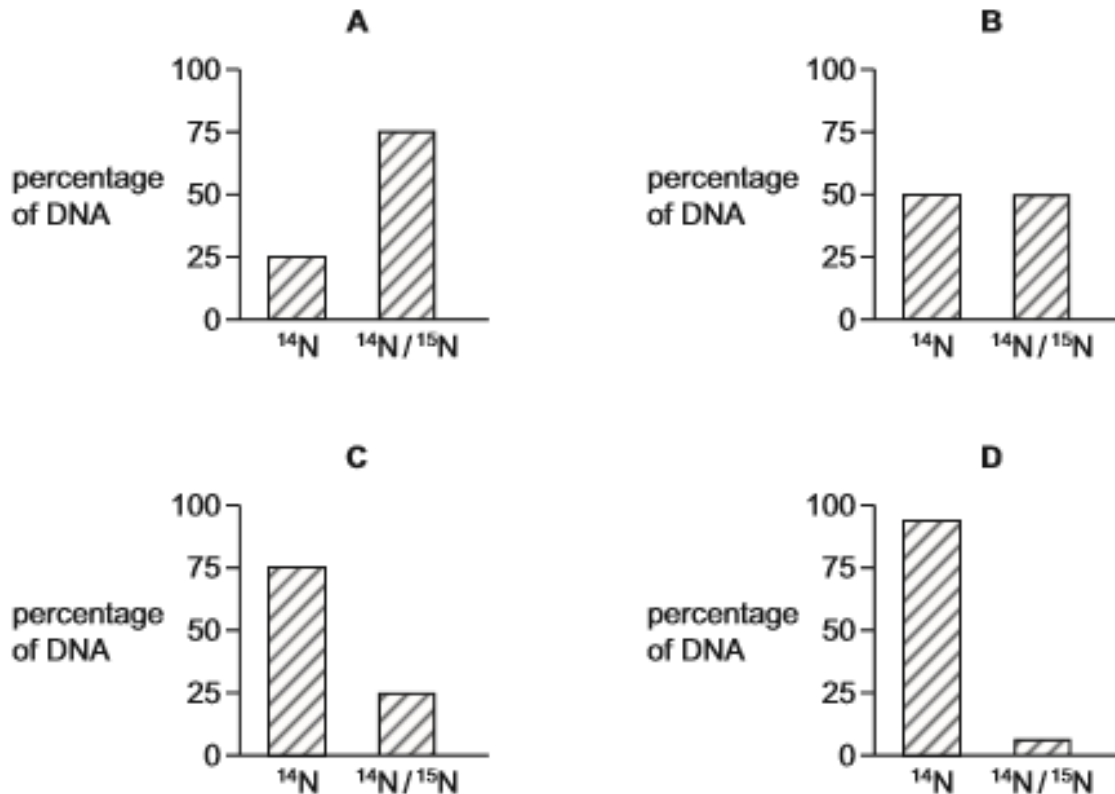
- Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.
- Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.
- Only mRNAs with polyadenine tailing can be translated.
- Polyadenine tailing stabilises mRNA and prevents it from being broken up.

- A** 1 and 2
B 1, 2 and 3
C 3 and 4
D 1, 2, 3 and 4

14. Bacteria were grown in a medium containing ^{15}N . After several generations, all the DNA contained ^{15}N . Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N . The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N .

The bacteria were allowed to divide two more times.

What would be the percentages of ^{14}N and ^{15}N labelled DNA in these bacteria?



15. A geneticist determines that a particular human disease is caused by a gene mutation. The mutant allele contains a substitution of cytosine to adenine at position 334. The DNA sequence for bases 301 to 351 from the non-template strand of the normal allele is shown.

5'- ATG TTA CGA GGT ATC ATA CGA ACG GAG CGC GAA CTA GTT ACT CCC ATA AAA - 3'

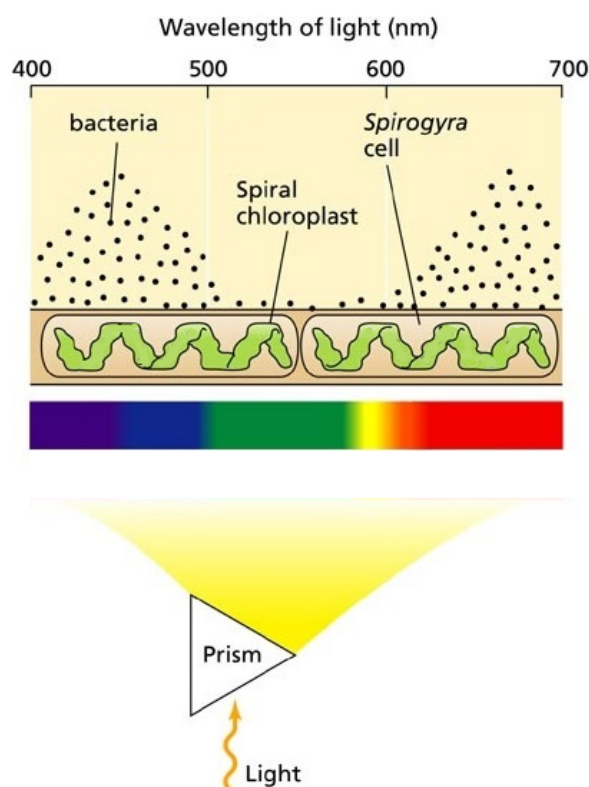
Which of the following statements best describes a consequence of this mutation?

- A A nonsense mutation has occurred resulting in no protein product being formed.
- B The mutant protein contains fewer amino acids than the normal protein.
- C A missense mutation has occurred resulting in a non-functional protein.
- D There is no change in length of amino acid sequence due to the mutation occurring outside of the coding region.

16. In 1882, the German botanist T.W. Engelmann performed an ingenious experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis for a suspension of alga *Spirogyra* (a filamentous microorganism containing long, spiral chloroplasts).

In his experiment, a prism was placed between the light source and the alga filament to produce and scatter all colours of the light across the alga filament evenly. Then, aerobic bacteria were added to the alga filament suspension. All the other variables were kept constant.

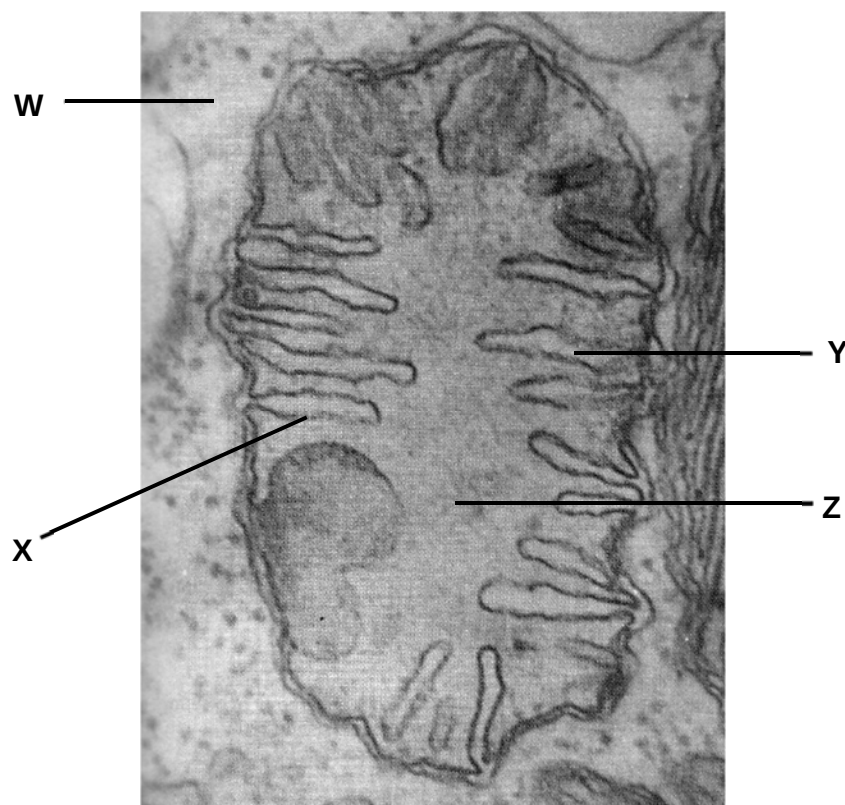
After exposure to light for a certain period of time, the bacteria were found to move towards and accumulate at specific lengths along the alga filament as shown below.



Which of the following conclusion(s) can be drawn from the above experiment.

- I The directional movement of the bacteria is due to oxygen released from *Spirogyra*.
 - II Green light is least absorbed, whereas red and blue wavelength of light is efficiently used for photosynthesis.
 - III NADPH is the reducing power that drives the formation of glyceraldehyde-3-phosphate.
- A** II only
B I and II only
C II and III only
D All of the above

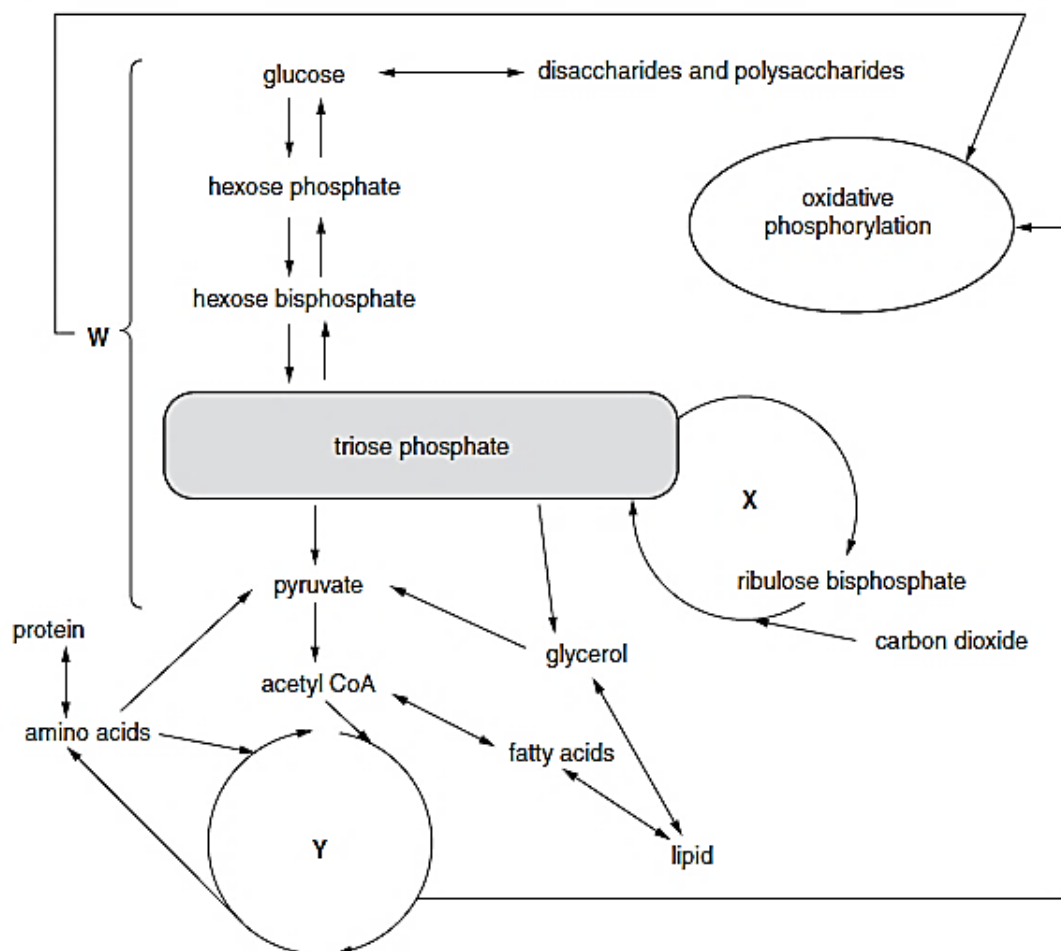
17. The figure below shows an electron micrograph of an organelle.



Which of the following correctly matches the following processes with the corresponding structures?

	Formation of pyruvate	Oxidative phosphorylation	Direction of diffusion of H^+ ions	Formation of reduced co-enzymes
A	W	X	$Z \rightarrow Y$	Z
B	Z	Y	$Z \rightarrow Y$	W and Z
C	Z	Y	$Y \rightarrow Z$	W
D	W	X	$Y \rightarrow Z$	W and Z

18. The diagram represents some of the reactions that take place in a leaf cell.



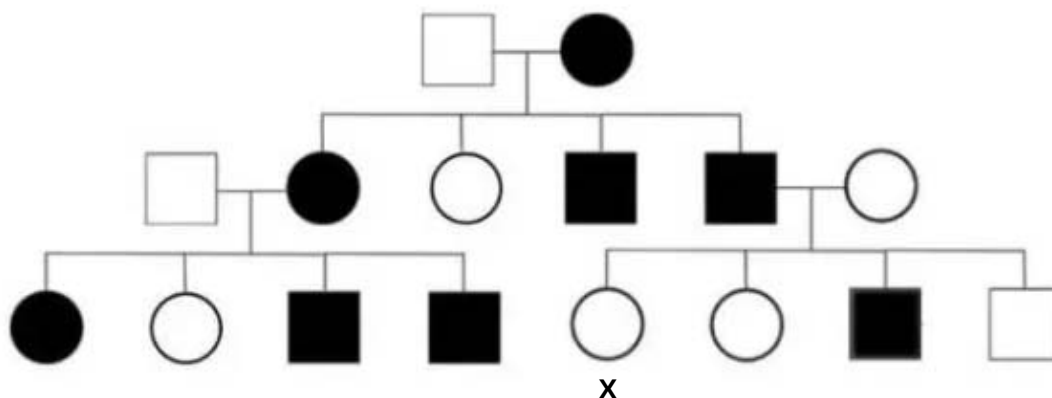
Which statement explains why the three reaction pathways **W**, **X** and **Y** are able to work independently of one another in the same leaf cell?

- A **X** occurs only in the absence of light unlike both **W** and **Y**.
- B **W**, **X** and **Y** are separated by membranes, allowing for the formation of separate proton gradients to synthesise ATP.
- C **W**, **X** and **Y** are separated by membranes, allowing for the maintenance of different conditions for enzymes to function.
- D **W**, **X** and **Y** are different pathways, each with their own biochemical reactions.

19. A metabolic poison inhibits liver enzymes that convert lactate to other compounds for metabolism. Which row illustrates the events that occur in an exercising athlete who accidentally consumed the metabolic poison?

	lactate accumulation	NAD production	ATP production	pH of blood
A	no	no	no	increased
B	no	yes	yes	decreased
C	yes	no	no	increased
D	yes	yes	yes	decreased

20. The pedigree below shows the inheritance of Marfan syndrome which affects connective tissues in the body.



Individual X is homozygous at the loci for the disease gene. What is the genetic basis of inheritance of the disease?

- A** Autosomal dominant
- B** Autosomal recessive
- C** Sex-linked dominant
- D** Sex-linked recessive

- 21.** A scientist was studying the inheritance of coat colour and tail shape in mice.

Coat colour is controlled by the *A/a* gene locus, where allele *A* codes for black coat colour, whereas tail shape is controlled by the *B/b* gene locus, where allele *B* codes for a kinked tail.

A black mouse with straight tail was crossed with an albino mouse with kinked tail. The cross produced the following offspring:

7 black mouse with kinked tail
7 black mouse with straight tail
8 albino mouse with kinked tail
5 albino mouse with straight tail

Based on the information above, what is the genotype of the parent mice that gave rise to the above results?

- A** *AA**bb* x *aa**BB*
- B** *Aabb* x *aaBb*
- C** *AaBb* x *aabb*
- D** *AA**bb* x *aa**BB*

22. Border collie dogs can have “stand up” ears, “flop down” ears or “mid way” ears. Ear type is controlled by a single gene.

A true breeding “stand up” ears Border collie was crossed with a true breeding “flop down” ears Border collie.



“stand up” ears



“flop down” ears

All the Border collies in the F_1 generation had “mid way” ears.

Based on the information given, which of the following statement(s) is/are valid.

- 1 The basis of inheritance for ear type is codominance, where the heterozygotes will have “mid way” ears.
- 2 The cross of two border collies with “mid way” ears will give rise to offspring with a phenotypic ratio of 1 : 2 : 1.
- 3 Ear type phenotype is controlled by one gene locus and three alleles.
- 4 The genotype of a border collie with “stand up” ears can only be determined by crossing with a border collie with “mid way” ears.

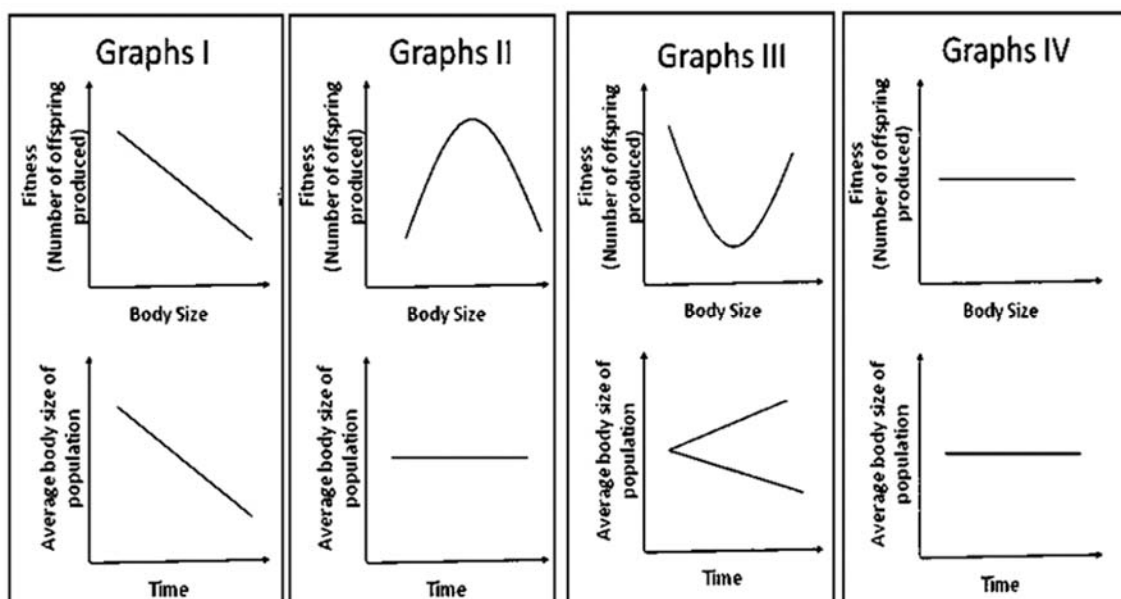
- A 2 only
- B 3 only
- C 1 and 2 only
- D 1, 2 and 4 only

23. Totipotency is demonstrated when _____.

- A cancer cells give rise to heterogeneous cell types
- B a stem cell can differentiate into placental cells and all cells in an organism
- C a hematopoietic stem cell differentiates into a lymphocyte
- D an embryonic stem cell divides and differentiates

- 24.** The use of induced pluripotent stem cells (iPSCs) overcome some of the following issues involving the use of other types of stem cells. Which issue below constitutes an ethical issue?
- I Embryonic stem cells are harvested from the blastocyst, which have the potential to develop into an adult and its use takes away a life.
 - II Stem cells are difficult to harvest from an individual as they are present in smaller proportions and are indistinguishable from non-stem cells.
 - III The use of embryonic stem cells in stem cell therapy may result in life threatening tissue rejection in the patient.
 - IV Consent on the use of embryonic stem cells is currently given by the parents but the potential being had no say.
- A** II and III
 - B** I and IV
 - C** I, III and IV
 - D** All of the above

25. The different forms of natural selection can be distinguished according to their effect on the body size of the pink salmon (*Onchorhynchus gorbuscha*).



Which of the following describes the correct form of natural selection for each of the following sets of graphs?

	Graphs I	Graphs II	Graphs III	Graphs IV
A	Disruptive selection	Directional selection	Stabilising selection	No selection
B	No selection	Stabilising selection	Directional selection	Disruptive selection
C	Directional selection	Stabilising selection	Disruptive selection	No selection
D	Directional selection	Disruptive selection	No selection	Stabilising selection

- 26.** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperms, and they selectively kill developing male embryos.

In the 1960s in Samoa, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

- I *Wolbachia* acts as a selective agent.
 - II The selective killing of male embryos is an example of artificial selection.
 - III When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
 - IV All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.
 - V The frequency of the dominant allele of the suppressor gene rises in the butterfly population.
-
- A** I and IV
 - B** II and III
 - C** I, III and V
 - D** II, IV and V

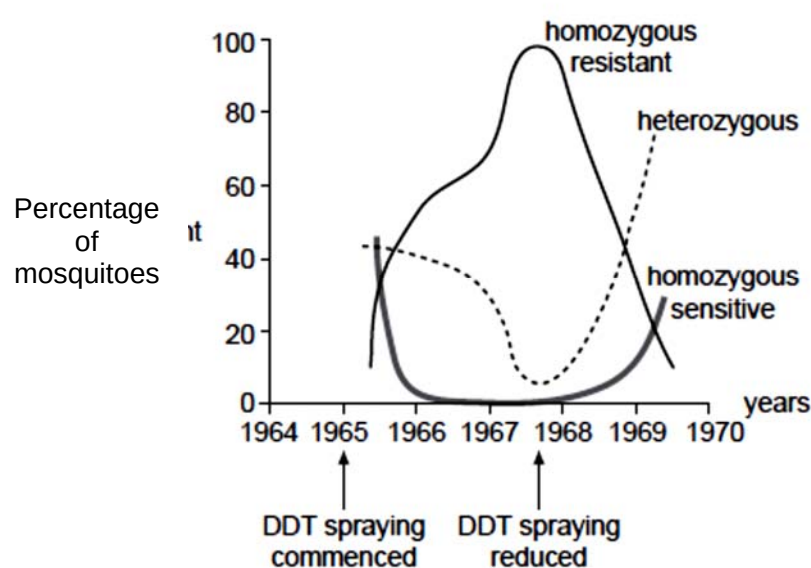
27. In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles.

allele 1 : resistant to DDT

allele 2 : sensitive to DDT

Over several years, genotypic frequencies were measured in a population of mosquito larvae.

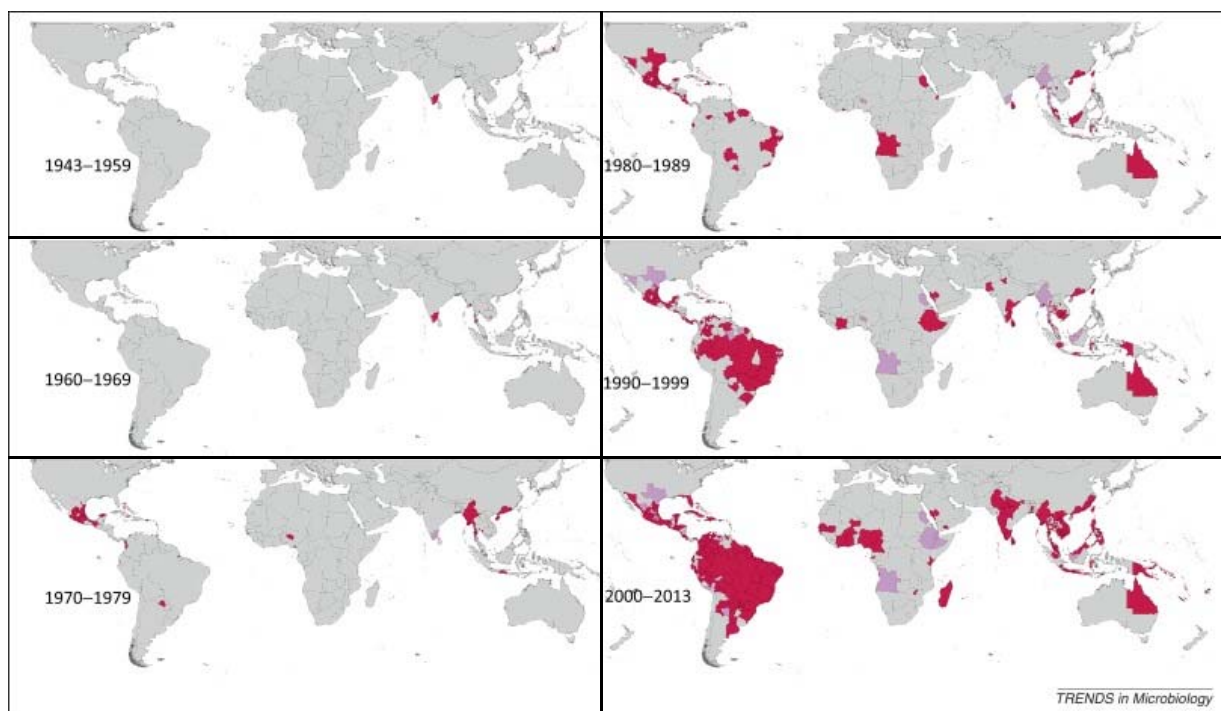
The graph below shows the results.



Analysis of the graph reveals that in the population, _____.

- A allele 1 confers a selective disadvantage in the absence of DDT.
 - B heterozygote advantage is demonstrated after DDT spraying is reduced.
 - C mutant allele 1 emerged as a result of the use of DDT in 1965.
 - D only one copy of allele 1 is required for resistant phenotype.
28. Which of the following describes a positive feedback loop concerning climate change?
- A Increased atmospheric temperature result in melting of sea ice which decreases the amount of sunlight reflected back into space
 - B Increased burning of fossil fuels increases atmospheric CO₂ concentration, enhancing the greenhouse effect
 - C Melting of glaciers causes an increase in sea levels
 - D Increase in atmospheric temperature causes many species to move towards increased altitudes to stay within their optimum temperature range

29. The diagram below shows the distribution of confirmed cases of dengue fever from 1943 to 2013.



Which of the following explain(s) the observed changes in distribution?

- I Increased global human traffic
- II Increased global temperatures allows mosquitoes to survive better at increased latitudes
- III Increased global temperatures allow mosquitoes carrying the dengue virus to move northwards
- IV Increased global temperatures increases the replication rate of the dengue virus in mosquitoes
- V Increased global temperatures reduces the replication rate of the dengue virus in humans

- A II and V only
- B II, III and IV only
- C I, II and IV only
- D All of the above

- 30.** In the recent decades, human activities have contributed significantly to the increase in greenhouse gases, resulting in global warming.

Which of the following are not examples of such human activities?

- A** Burning of fossil fuels
- B** Conversion of land for agriculture
- C** Increase consumption of meat
- D** Overfishing

- End of Paper -